

CHE374 — Economic Analysis and Decision Making

Course Notes

Bonnie Zhuang

Fall 2026

Contents

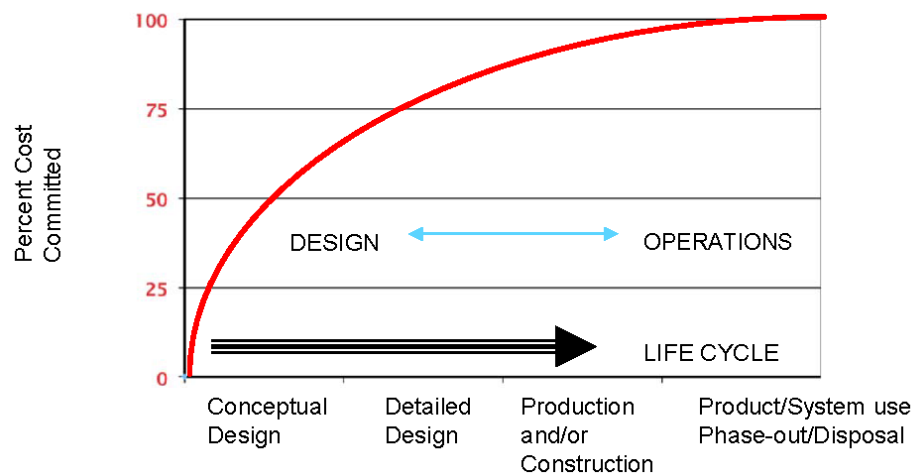
Week 1 - Introduction and Time Value of Money	3
1.1 - Interest Rate	3
1.2 - Simple and Compound Interest	5
1.3 - Subperiod Compounding & Effective Interest Rates	7
Week 2 - Cash-flow Analysis	10
2.1 - Cash-flow Diagrams	10
2.2 - Categories of Regular Cash-flows	12
2.3 - Cash-flow Equivalence	13
2.4 - Cash-flow Factors	15
2.5 - Worked Examples	19
Week 3 - Financial Instruments	22
3.1 - Mortgages	22
3.2 - Bonds	23
3.3 - Bond Examples	27
Week 4 - Risk, Reward and Arbitrage	28
4.1 - Capital Asset Pricing Model	28
4.2 - Arbitrage	32

Week 1 - Introduction and Time Value of Money

1.1 - Interest Rate

Engineering Project Life Cycles

- All engineering projects have a life cycle:
 - Conceptualization → detailed economic analysis → go / no go decision → design → build → utilize / produce → phase out / dispose
- Most of the project's cost is **committed** early, during conceptual and detailed design, even though the money is mostly spent later (production, operation)
- We **invest** in projects to gain benefits (economic and societal)



Most cost is committed early, during design, before it is actually spent

Time Value of Money

- Borrowing (renting) money has a cost for the **borrower**, thus lending money *should* create value for the lender (investor/creditor)
- Costs and benefits occur in different time periods
- The cost/benefit today to the future is compared by **interest**

Example

You borrow \$10 from a lender today and return it a year later.

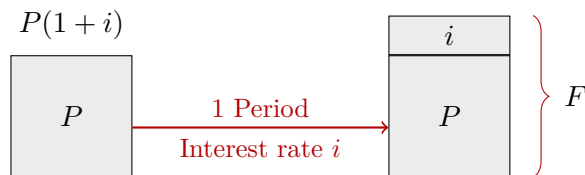
- You must return the \$10 borrowed ⇒ the **principal**
- The lender charges an extra amount, e.g. \$1 ⇒ the **interest**
- The **interest rate** is interest/principal:

$$i = \frac{1}{10} = 10\%$$

Present and Future Worth

P = present worth/**Principal** F = future worth i = interest rate

$$F = P + Pi \quad F = P(1 + i)$$



- A higher payoff (F) is expected as i increases
- For a given F with increased i , P decreases

Interest Rate

- Expressed as %
- Vary widely over time and different types of investments.
 - A **T-Bill** (Treasury Bill, short term loan to the Bank of Canada) will earn the lowest rate of return, as it's considered "risk-free"
 - Canadian and provincial bonds pay typically slightly higher interest rates
 - Personal/small business loans are much riskier and earn significantly higher interest rates

Factors that Affect Interest Rates

- **Inflation:** prices generally tend to increase over time. Higher expected inflation generally leads to higher interest rates
- **Credit (default) Risk:** associated with the borrower being able to pay back interest and principal. Higher risk of default $\Rightarrow$ higher interest rates
- **Liquidity Risk:** associated with being able to access the invested funds during the investment. The less liquid the investment (greater the risk), the higher the interest rate
- **Maturity Risk:** associated with the time horizon. Longer the maturity, longer payoff, increases credit (and potential liquidity) risk $\Rightarrow$ higher interest rates

Generally, as **risk** increases, the **interest rate** increases.

Example

What would the impact be expected on the interest rate associated with a large new mining project under the following scenarios, and how would the current value be impacted?

- The expected time to complete the project construction increases
Interest rate will increase as the maturity risk is directly increased, raising risks of liquidity and credit. The present value would then decrease, from both the interest rate increasing and the time horizon increasing.
- The estimate amount of ore available to be mined increases
The future cash flow would increase, thus the value of the project will increase, decreasing risk. Credit risk will drop but not others.

Summary

- **Interest** is money earned by investors for allowing others to use their money
- The **interest rate** is a rate at which interest is earned.
- The interest rate is determined by **risks** associated with the investment.

1.2 - Simple and Compound Interest

Present and Future Worth

P = amount of money today (principal) F_N = future amount in year N

i = interest rate I = interest amount

$$F = P + I \quad F_1 = P + Pi$$

Example GIC

You deposit \$5,000 in the bank for 1 year as a **Guaranteed Investment Certificate (GIC)**, i.e. you *lent* the bank the money for a fixed time period. After 1 year the bank pays 10% interest.

You get $I = 5000(0.10) = \$500$ as “rent” for lending the \$5,000, and the \$5,000 is also returned.

- This is a **simple** interest loan: you receive the money when the GIC **matures**
- **Matures**: the borrower must pay the loan back

Simple vs Compound Interest

- **Simple** applies only to the original principal (GIC, aka Certificate of Deposit (CD) in the U.S.)
- **Compound** interest applies to the principal AND to all interest already accrued (mortgages, credit card debt...)
- Assume compound unless stated otherwise

Simple Interest

Beginning of Period	Amount Lent	Interest Amount	Amount Owed at Period End
1	P	Pi	$P + Pi$
2	$P + Pi$	Pi	$P + 2Pi$
3	$P + 2Pi$	Pi	$P + 3Pi$
...	...	...	...
N	$P + (N - 1)Pi$	Pi	$P + NPi = P(1 + Ni)$

Compound Interest

Beginning of Period	Amount Lent	Interest Amount	Amount Owed at Period End
1	P	Pi	$P(1 + i)$
2	$P(1 + i)$	$P(1 + i)i$	$P(1 + i)^2$
3	$P(1 + i)^2$	$P(1 + i)^2i$	$P(1 + i)^3$
...	...	...	...
N	$P(1 + i)^{N-1}$	$P(1 + i)^{N-1}i$	$P(1 + i)^N$

Example Simple Interest

If you borrow \$100 for 3 years at 10% annual interest (simple interest), how much interest will you pay, and how much will you owe after 3 years?

$$P = 100 \quad i = 10\% \quad N = 3$$

$$I = NPi = 3(100)(0.10) = \$30$$

$$F_3 = P + NPi = 100 + 30 = \$130$$

Example Compound Interest

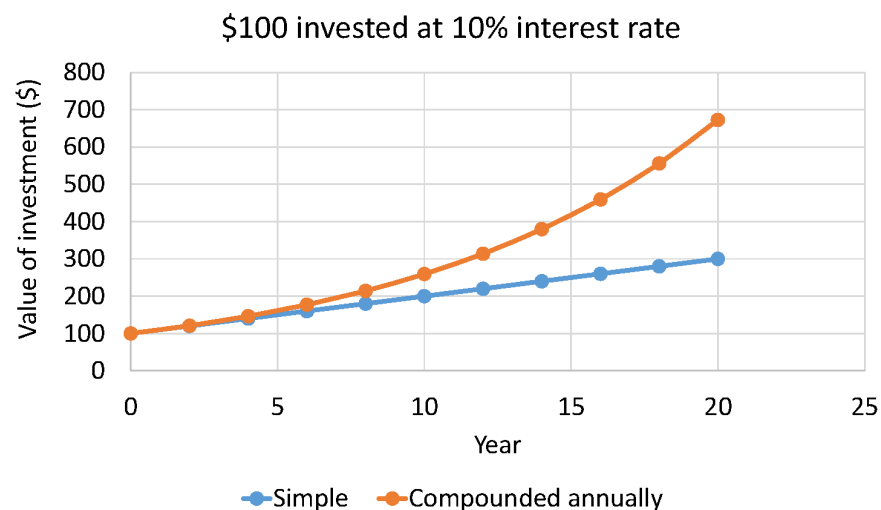
If you borrow \$100 for 3 years at 10% annual interest (compound interest), how much interest will you pay, and how much will you owe after 3 years?

$$F_3 = P(1 + i)^N = 100(1.10)^3 = \$133.10$$

$$I = F_3 - P = 133.10 - 100 = \$33.10$$

The extra \$3.10 compared to simple interest is interest earned on interest.

Simple Interest is Linear, Compound Interest is Exponential



\$100 invested at 10%: \$300 (simple) vs. \$673 (compound) after 20 years

Example The Power of Compounding

The Dutch West India Company (supposedly) paid \$24 to purchase Manhattan Island in 1626. If the company had invested the \$24 in a savings account that earned 7% interest per year, how much would it be worth today (2020, $N = 394$)?

$$\text{Simple: } F = 24 + 394(24)(0.07) \approx \$686$$

$$\text{Compound: } F = 24(1.07)^{394} \approx \$9,066,082,143,625$$

Year	Simple (7%)	Compound (7%)
1626	\$24	\$24
1676	\$108	\$707
1726	\$192	\$20,825
1776	\$276	\$613,448
1826	\$360	\$18,070,359
1876	\$444	\$532,299,016
1926	\$528	\$15,679,945,443
1976	\$612	\$461,884,545,916
2020	\$686	\$9,066,082,143,625

Summary

- Simple Interest: $F_N = P + NPi$
- Compound Interest: $F_N = P(1 + i)^N$
- Remember: **interest rate** = interest/principal

1.3 - Subperiod Compounding & Effective Interest Rates

Compound Interest

- Recall: **compound** interest applies to the principal AND to all interest already accrued
- “To compound” (verb) is to reinvest the principal PLUS the interest and earn interest on both:

$$F = P(1 + i)^N$$

Compound Multiple Times per Year

- **r** : nominal interest rate (usually for 1 year)
- **m** : times compounded in a year, aka **subperiods**
- **i_s** : subperiod interest rate

$$i_s = \frac{r}{m}$$

Example

You lend \$100. The nominal interest rate is 6%.

What is the total amount in the account at the end of the year if the compounding period is

$$P = 100 \quad r = 6\%$$

a) 1 year:

$$\begin{aligned} F &= P(1 + r) \\ &= 100(1 + 0.06) \\ &= 106 \end{aligned}$$

b) 3 months:

$$m = 4 \Rightarrow i_s = \frac{r}{m} = \frac{0.06}{4} = 0.015$$

$$\begin{aligned} F &= P(1 + i_s)^m \\ &= 100(1 + 0.015)^4 \\ &= 106.14 \end{aligned}$$

Example Reflection

True or false: for the same nominal interest rate, the more frequently you compound, the more you earn at the end of the year.

True. Each extra compounding lets earlier interest start earning interest sooner (\$106 vs. \$106.14 above). The gain shrinks as m grows, though, approaching a finite limit (continuous compounding).

Effective Interest Rate

- How do we compare investments with different compounding periods?
- **i_e : effective interest rate**, the equivalent interest rate of an account that is compounded just *once* over the stated time period (usually 1 year)
- Set one year of subperiod compounding equal to one compounding at i_e :

$$P(1 + i_e) = P \left(1 + \frac{r}{m} \right)^m \Rightarrow \boxed{i_e = \left(1 + \frac{r}{m} \right)^m - 1}$$

Example

Which account earns you more money after 1 year?

- Option A: $r = 6\%$, compounded monthly
- Option B: $r = 6.3\%$, compounded once a year

$$\text{A: } i_e = \left(1 + \frac{0.06}{12} \right)^{12} - 1 = (1.005)^{12} - 1 = 6.17\%$$

$$\text{B: } i_e = r = 6.3\% \quad (m = 1)$$

Option B earns more, since $6.3\% > 6.17\%$. Compare the effective rates, not the nominal ones.

Continuous Compounding

- As the compounding period shortens (m increases), i_e increases

- As the compounding period becomes infinitesimally small, i_e approaches a finite limit $\Rightarrow$ **continuous compounding**

$$i_e = \lim_{m \rightarrow \infty} \left(1 + \frac{r}{m}\right)^m - 1 = e^r - 1$$

- e.g. $r = 6\%$: $i_e = e^{0.06} - 1 = 6.18\%$, only slightly more than monthly (6.17%)

Summary

- Know the difference between **nominal** and **effective** interest rates, and how to convert between them
 1. Start with the nominal interest rate r for a stated time period (usually 1 year)
 2. Compound multiple (m) times within that time period, at $i_s = r/m$
 3. Effective interest rate: $i_e = (1 + r/m)^m - 1$
 4. Continuous compounding: $i_e = e^r - 1$

Week 2 - Cash-flow Analysis

2.1 - Cash-flow Diagrams

Remark (Project Life-Cycle Cost Categories). • **Start: First (capital) cost** — expense to build/buy and install.

- **During: Revenues (sales)** — receipts from sale of products/services; **O&M costs** — expenses incurred on a regular basis (electricity, labour, repairs); **Overhaul** — a major capital expenditure partway through the asset's life.
- **End: Salvage value** — net receipt from sale/disposal at the end of the asset's *useful* life; **Scrap value** — value of the materials at the end of the asset's *physical* life; **Disposal costs** — cost to dispose of waste (removal, shipping, tipping fees, etc.).

Definition - Cash-flow Table

A table showing the “money consequences” of a situation and the timing of the cash-flows: for each point in time, the activity and the resulting cash-flow. Costs/benefits occurring at the start, during, or end of a project are together called the **project life-cycle costs**.

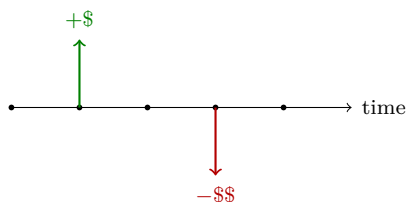
Example Car Ownership Cash-flow Table

Year	Cash-flow	Activity
Start of Year 1	−\$24,500	Car purchased now
End of Year 1	−\$1,500	Service cost
End of Year 2	−\$1,500	Service cost
End of Year 3	−\$1,500	Service cost
End of Year 4	−\$1,500, +\$5,000	Service cost & car sold for \$5,000

The purchase is the first cost (at $t = 0$), service is an O&M cost, and the \$5,000 sale is the salvage value.

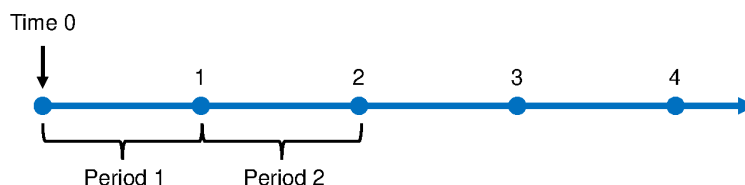
Definition - Cash-flow Diagram

A simple graph summarizing the **timing** and **magnitude** of cash-flows. The x -axis is discrete time periods; the (implicit) y -axis is the size and direction of each cash-flow, drawn as a vertical arrow: an **up** arrow is a receipt (inflow, +\$), a **down** arrow is a disbursement (outflow, −\$).



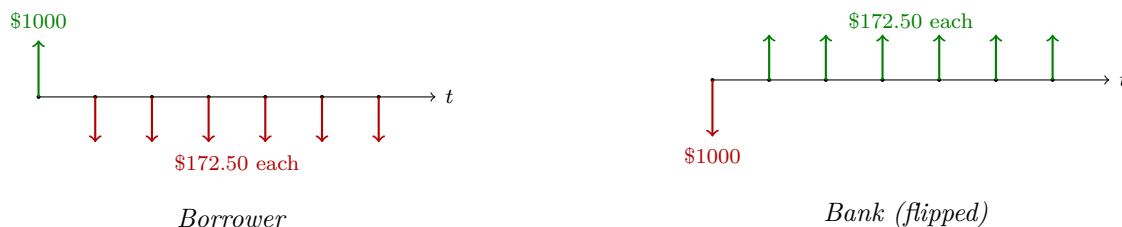
$Up = \text{inflow (receipt)}; \text{down} = \text{outflow (disbursement)}$

- Remark* (Diagram Conventions). • The end of one period is the beginning of the next: time 0 is the start of period 1, time 1 is the end of period 1 and the start of period 2, and N periods from now is the end of period N / start of period $N + 1$.
- Unless stated otherwise: interest compounds once per period, and cash-flows occur at the **end** of the period.
 - Cash-flows occurring *during* a period are **summed** and accounted for at the **end** of that period.
 - First costs are typically accounted for at **time 0**.



Example Loan Repayment Diagram

\$1,000 is borrowed on January 1 and repaid in 6 monthly installments of \$172.50 ($i = 1\%$ per month). The time line's origin is the start of the analysis (time 0 = January 1, intervals are monthly). From the **borrower's** perspective: an inflow of \$1,000 at $t = 0$, and outflows of \$172.50 at $t = 1, \dots, 6$. From the **bank's** perspective the diagram flips: an outflow of \$1,000 at $t = 0$, and inflows of \$172.50 at $t = 1, \dots, 6$.



Example Reflection: Young Professional

Construct a cash-flow diagram for one month (assume 4 weeks per month):

- Monthly income: \$2,200 (end of each month)
- Rent incl. utilities: \$700 (end of each month)
- Weekly food and entertainment: \$120
- Telephone bill: \$40 (end of the first week of each month)
- Credit card purchases: \$300 (end of the second week of each month)

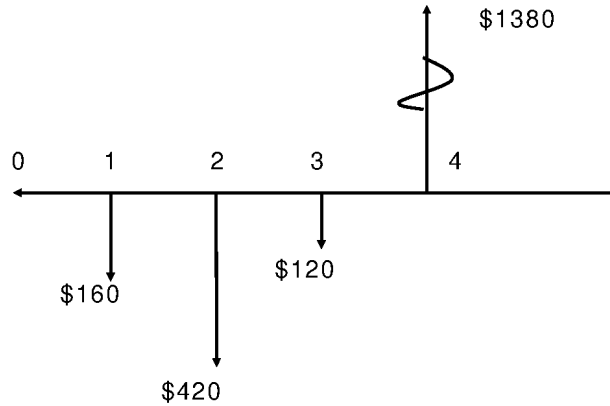
Use weeks as periods and net everything at the end of each week:

$$\text{Week 1 : } -120 - 40 = -\$160$$

$$\text{Week 2 : } -120 - 300 = -\$420$$

$$\text{Week 3 : } -\$120$$

$$\text{Week 4 : } 2200 - 700 - 120 = +\$1,380$$



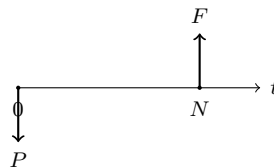
2.2 - Categories of Regular Cash-flows

Five Types of (Regular) Cash-flows

1. Single payments or receipts
2. Perpetuity
3. Annuity
4. Arithmetic gradient
5. Geometric gradient

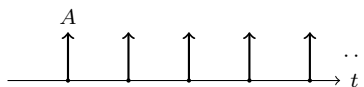
Definition - **Single Payment/Receipt**

A single cash-flow P (now) or F (in the future), occurring only once, N periods apart.



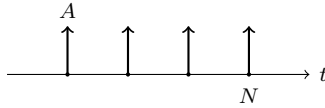
Definition - **Perpetuity**

A constant cash-flow A occurring every period, forever (an *infinite annuity*).



Definition - **Annuity**

A constant (uniform) cash-flow A occurring every period, for a *finite* number of periods N .

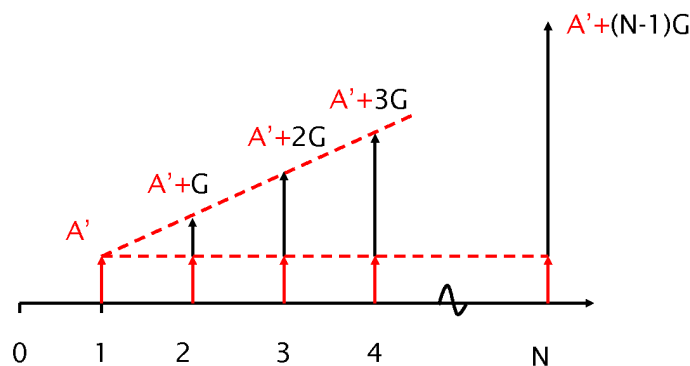


Definition - Arithmetic Gradient

A cash-flow that grows (or shrinks) by the same *amount* G each period, starting from a base A' at $t = 1$:

$$A', A' + G, A' + 2G, \dots, A' + (N - 1)G.$$

It splits into an annuity of A' (red, below) plus a pure gradient $G, 2G, \dots$ on top (black).

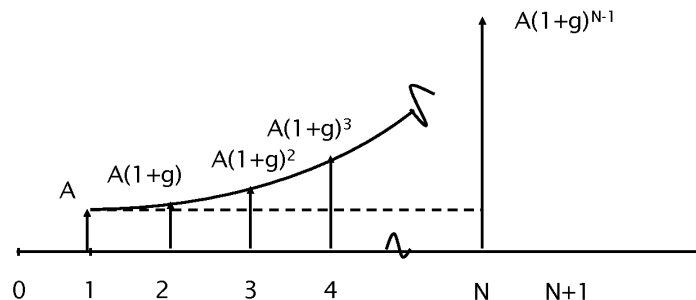


Definition - Geometric Gradient

A cash-flow that grows (or shrinks) by the same *percentage* g each period, starting from A at $t = 1$:

$$A, A(1 + g), A(1 + g)^2, \dots, A(1 + g)^{N-1}.$$

There must be a cash-flow at the end of period 1, otherwise there is nothing for the percentage to grow from.



2.3 - Cash-flow Equivalence

Three Concepts of Equivalence

- **Mathematical**: consequence of the mathematical relationship between time and money
- **Market**: consequence of the ability to exchange one cash-flow for another at zero cost
- **Decisional**: due to indifference on the part of the decision maker among available choices

Definition - Mathematical Equivalence

A consequence of the mathematical relationship between time and money. Two cash-flows P_t (at time t) and F_{t+N} (at time $t + N$) are equivalent at rate i if

$$F_{t+N} = P_t(1+i)^N \iff P_t = \frac{F_{t+N}}{(1+i)^N}.$$

If a further cash-flow F_{t+N+M} is also equivalent to P_t , then

$$F_{t+N+M} = P_t(1+i)^{N+M} = F_{t+N}(1+i)^M \iff P_t = \frac{F_{t+N+M}}{(1+i)^{N+M}},$$

i.e. equivalent amounts compound/discount consistently with each other.



Definition - Market Equivalence

A consequence of being able to exchange one cash-flow for another at *zero cost*. Exchanging a future cash-flow for a present one is **borrowing**; giving up a present cash-flow for a future one is **lending/investing**. In real markets, borrowing and lending rates usually differ.

Example T-bill Market Equivalence

A T-bill is a short-term government debt obligation (maturity ≤ 1 year), usually sold in \$1,000 denominations. A \$1,000 T-bill maturing in 6 months currently sells for \$990.10: some holders are willing to sell at this price and others to buy. The market has agreed this corresponds to an interest rate of 1% over 6 months. Confirm:

$$i = \frac{1000}{990.10} - 1 = 0.0100 = 1\%$$

Definition - Decisional Equivalence

Two cash-flows P_t and F_{t+N} are equivalent for a decision maker if they are *indifferent* between them. Here the interest rate is not known a priori — instead, an **implied** rate can be backed out from the decision that the two are equivalent.

Example Implied Interest Rate

A manufacturer needs 1,000 kg of aluminium on August 15 but is offered 1,100 kg on August 22 instead (same price, paid at the end of August).

$$i = \frac{1100}{1000} - 1 = 10\% \text{ for one week}$$

The real question is whether the extra 100 kg is worth the cost of the one-week delay to this decision maker. If they are indifferent, 10%/week is their implied rate.

Remark. Financial analysis uses **mathematical** and **market** equivalence to decide whether to invest or not invest.

2.4 - Cash-flow Factors

Remark (Standing Assumptions). Unless otherwise noted:

1. Interest compounds once per period.
2. Cash-flow occurs at the end of the period.
3. Time 0 is period 0, i.e. the start of period 1.
4. All periods are the same length.

Definition - Cash-flow Symbols

P : a single cash-flow today. F : a single cash-flow in the future. A : a finite series of equal cash-flows at regular intervals (annuity). G : a cash-flow growing by a fixed *amount* each period (arithmetic gradient) or fixed *percentage* g each period (geometric gradient, “Geom”).

Definition - Factor Notation

$(X/Y, i, N)$ converts a cash-flow of type Y into the equivalent amount of type X at rate i over N periods: $X = Y \cdot (X/Y, i, N)$. E.g. $F = P(F/P, i, N)$; for geometric gradients, $P = G(P/\text{Geom}, i, g, N)$.

Factor	Name
$(F/P, i, N)$	Compound amount factor
$(P/F, i, N)$	Present worth factor
$(A/F, i, N)$	Sinking fund factor
$(F/A, i, N)$	Series compound amount factor
$(A/P, i, N)$	Capital recovery factor
$(P/A, i, N)$	Series present worth factor
$(P/G, i, N)$	Arithmetic gradient to present worth
$(P/\text{Geom}, i, g, N)$	Geometric gradient to present worth

Remark (Relating Factors). Factors invert: since $P = A(P/A, i, N)$,

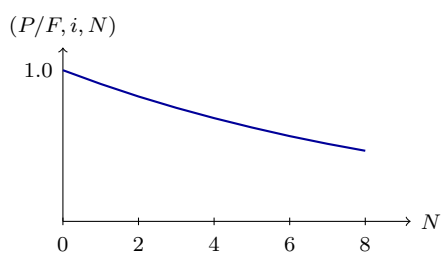
$$A = \frac{P}{(P/A, i, N)} = P(A/P, i, N) \Rightarrow (A/P, i, N) = \frac{1}{(P/A, i, N)}.$$

The same trick relates any pair of factors.

Definition - Single-Payment Factors

P and F are related by compounding one period at a time, $F_n = F_{n-1}(1+i)$, so $F = P(1+i)^N$:

$$(F/P, i, N) = (1+i)^N, \quad (P/F, i, N) = (1+i)^{-N}.$$



$(1+i)^{-N}$ decays toward 0 as N grows (shown for $i = 10\%$)

Example F/P

If you had \$2,000 now and invested it at 10%, how much would it be worth in 8 years?

$$F = 2000(F/P, 10\%, 8) = 2000(1.10)^8 = 2000(2.1436) = \text{\textcolor{blue}{\$4,287.18}}$$

Example P/F: Solving for i

You invest \$1,100 today and expect \$2,000 in 8 years. What is the interest rate?

$$1100 = 2000(P/F, i, 8) \Rightarrow (1+i)^8 = \frac{2000}{1100} = 1.8182 \Rightarrow i = 1.8182^{1/8} - 1 = \text{\textcolor{blue}{7.76\%}}$$

Example Present Value of a Perpetuity

Invest \$100 and get paid 5% interest in cash every year, forever. Each year the \$5 of interest is paid out, so the balance never changes:

Year	Interest	Cash-flow	Amount in bank at year end
0	—	−100	100
1	5	+5	100
2	5	+5	100
3	5	+5	100
⋮	⋮	⋮	⋮

So \$100 today is equivalent to \$5/year forever, i.e. $P = A/i = 5/0.05 = 100$. In general, a constant amount A paid forever has a finite present value:

$$P = \frac{A}{1+i} + \frac{A}{(1+i)^2} + \cdots = \boxed{\frac{A}{i}}.$$

Definition - Present Worth of an Annuity

An annuity of A for N periods is a perpetuity starting now minus a perpetuity starting at N (worth A/i at time N , discounted back):

$$(P/A, i, N) = \frac{1}{i} - \frac{1}{i(1+i)^N} = \frac{(1+i)^N - 1}{i(1+i)^N}.$$

Combining with (F/P) and inverting gives the rest of the uniform-series factors:

$$(F/A, i, N) = \frac{(1+i)^N - 1}{i}, \quad (A/F, i, N) = \frac{i}{(1+i)^N - 1}, \quad (A/P, i, N) = \frac{i(1+i)^N}{(1+i)^N - 1}.$$

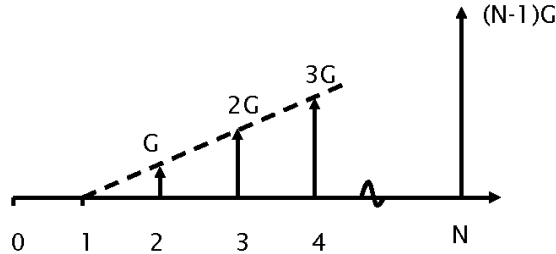
Definition - Present Worth of an Arithmetic Gradient

For a gradient contributing $G, 2G, \dots, (N-1)G$ starting at $t = 2$ (no cash-flow at $t = 0$ or $t = 1$):

$$(P/G, i, N) = \frac{1}{i^2} \left(1 - \frac{1+iN}{(1+i)^N} \right).$$

A series with initial amount A growing by G each period splits into an annuity plus a gradient:

$$P = A(P/A, i, N) + G(P/G, i, N).$$



The pure gradient: nothing at $t = 0, 1$; G at $t = 2$, ..., $(N-1)G$ at $t = N$

Remark (Sign Combinations for Arithmetic Gradients). Besides $G = 0$, four cases: $A > 0, G > 0$ (positive, increasing); $A > 0, G < 0$ (positive, decreasing); $A < 0, G > 0$ (negative, becoming less so); $A < 0, G < 0$ (negative, becoming more so).

Definition - Present Worth of a Geometric Gradient

For a series $A, A(1+g), A(1+g)^2, \dots$ starting at $t = 1$:

$$(P/\text{Geom}, i, g, N) = \frac{1 - \left(\frac{1+g}{1+i}\right)^N}{i - g} = \frac{(P/A, i_o, N)}{1+g}, \quad i_o = \frac{1+i}{1+g} - 1.$$

Remark. Regardless of the pattern, present value always follows from discounting each cash-flow individually and summing. Each period a cash-flow is moved forward, multiply by $(1 + i)$; each period back, divide by $(1 + i)$:

$$P = \frac{F_1}{1+i} + \frac{F_2}{(1+i)^2} + \cdots = \sum_t \frac{F_t}{(1+i)^t}.$$

Factor Table ($i = 10\%$)

$i = 10\%$ Discrete Compounding, Discrete Cash Flows							
	SINGLE PAYMENT		UNIFORM SERIES				Arithmetic Gradient Series Factor
	Compound Amount Factor	Present Worth Factor	Sinking Fund Factor	Uniform Series Factor	Capital Recovery Factor	Series Present Worth Factor	
N	$(F/P, i, N)$	$(P/F, i, N)$	$(A/F, i, N)$	$(F/A, i, N)$	$(A/P, i, N)$	$(P/A, i, N)$	$(A/G, i, N)$
1	1.1000	0.90909	1.0000	1.0000	1.1000	0.90909	0.00000
2	1.2100	0.82645	0.47619	2.1000	0.57619	1.7355	0.47619
3	1.3310	0.75131	0.30211	3.3100	0.40211	2.4869	0.93656
4	1.4641	0.68301	0.21547	4.6410	0.31547	3.1699	1.3812
5	1.6105	0.62092	0.16380	6.1051	0.26380	3.7908	1.8101
6	1.7716	0.56447	0.12961	7.7156	0.22961	4.3553	2.2236
7	1.9487	0.51316	0.10541	9.4872	0.20541	4.8684	2.6216
8	2.1436	0.46651	0.08744	11.436	0.18744	5.3349	3.0045
9	2.3579	0.42410	0.07364	13.579	0.17364	5.7590	3.3724
10	2.5937	0.38554	0.06275	15.937	0.16275	6.1446	3.7255
11	2.8531	0.35049	0.05396	18.531	0.15396	6.4951	4.0641
12	3.1384	0.31863	0.04676	21.384	0.14676	6.8137	4.3884
13	3.4523	0.28966	0.04078	24.523	0.14078	7.1034	4.6988
14	3.7975	0.26333	0.03575	27.975	0.13575	7.3667	4.9955
15	4.1772	0.23939	0.03147	31.772	0.13147	7.6061	5.2789
16	4.5950	0.21763	0.02782	35.950	0.12782	7.8237	5.5493
17	5.0545	0.19784	0.02466	40.545	0.12466	8.0216	5.8071
18	5.5599	0.17986	0.02193	45.599	0.12193	8.2014	6.0526
19	6.1159	0.16351	0.01955	51.159	0.11955	8.3649	6.2861
20	6.7275	0.14864	0.01746	57.275	0.11746	8.5136	6.5081

Example Reflection: Lump Sums vs. Annuity

Which option do you prefer if interest is 8%? (a) \$20,000 now and \$50,000 at the end of 10 years, or (b) ten equal annual receipts of \$7,000?

$$P_a = 20,000 + 50,000(P/F, 8\%, 10) = 20,000 + 50,000(0.46319) = \text{\textcolor{blue}{\$43,160}}$$

$$P_b = 7,000(P/A, 8\%, 10) = 7,000(6.7101) = \text{\textcolor{blue}{\$46,971}}$$

Prefer (b): it has the higher present value.

Example Reflection: Annuity at a Lower Rate

\$1/year for 5 years has a present value of \$3.7908 at 10%. Which is a likely value at 9%? (a) 3.8897 (b) 5.0000 (c) 3.7908 (d) 2.3262

(a) 3.8897. A lower rate discounts less, so PV rises above 3.7908, but it must stay below the undiscounted total of \$5. Check: $(P/A, 9\%, 5) = 3.8897$.

2.5 - Worked Examples

Example Investment Decision

An investor can buy a tract of land that will be worth \$10,000 in six years. If the rate of return on land is 8%, how much should the investor be willing to pay now?

$$P = 10,000(P/F, 8\%, 6) = \frac{10,000}{(1.08)^6} = \$6,301.70$$

Example The Magic of Compounding

At age 20 you save and invest \$1.00 each day (\$365/year) until age 80, at 10% annual interest.

$$F = 365(F/A, 10\%, 60) = 365 \cdot \frac{(1.10)^{60} - 1}{0.10} \approx \$1,107,708$$

You become a millionaire.

Example F and P with Subperiod Compounding

How much is accumulated over 5 years in each savings plan?

- \$2,000 today at 10% compounded semi-annually: $i_s = 5\%$, $N = 10$

$$F = 2000(1.05)^{10} = \$3,257.79$$

- \$1,000 today at 12% compounded monthly: $i_s = 1\%$, $N = 60$

$$F = 1000(1.01)^{60} = \$1,816.70$$

How much must you invest today to get \$2,000 in 5 years at 8% compounded quarterly? $i_s = 2\%$, $N = 20$:

$$P = 2000(P/F, 2\%, 20) = \frac{2000}{(1.02)^{20}} = \$1,345.94$$

Example Mortgage/Amortization

A \$300,000 mortgage at a nominal 9% (mortgages are quoted with semi-annual compounding) is amortized over 15 years. What is the monthly payment?

Semi-annual rate = 4.5%. The equivalent monthly rate satisfies $(1 + i_m)^6 = 1.045$:

$$i_m = 1.045^{1/6} - 1 = 0.7363\%, \quad N = 15 \times 12 = 180$$

$$A = 300,000(A/P, 0.7363\%, 180) = \$3,013.56 \text{ per month}$$

That is about \$36,163/year, over 70% of a \$50k starting salary (before tax), so it doesn't fit the budget.

Example Sinking Fund

A “targeted” F calculation: save up to pay a future expenditure (e.g. replacing a machine). A roof must be replaced in 10 years for an estimated \$28,000. What equal end-of-year deposits at 8% are

needed?

$$A = 28,000(A/F, 8\%, 10) = 28,000 \cdot \frac{0.08}{(1.08)^{10} - 1} = \$1,932.83 \text{ per year}$$

Example Retirement Planning

You are 20 and want \$1,000,000 at retirement at 65 (45 years). What equal end-of-year savings are needed at 7%? At 10%?

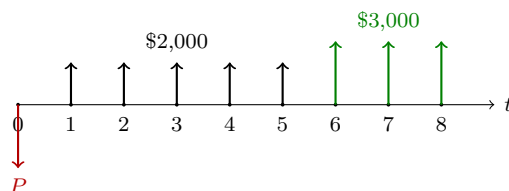
$$A_{7\%} = 1,000,000(A/F, 7\%, 45) = \$3,499.57 \text{ per year}$$

$$A_{10\%} = 1,000,000(A/F, 10\%, 45) = \$1,391.00 \text{ per year}$$

Example Uneven Annuity Segments

Claudia wants to deposit P now so that she can withdraw \$2,000/year for 5 years and then \$3,000/year for the next 3 years, at 8%. Discount the second annuity back to $t = 5$, then to $t = 0$:

$$\begin{aligned} P &= 2,000(P/A, 8\%, 5) + 3,000(P/A, 8\%, 3)(P/F, 8\%, 5) \\ &= 2,000(3.9927) + 3,000(2.5771)(0.68058) \\ &= 7,985.42 + 5,261.79 = \$13,247.21 \end{aligned}$$



Example Present Value with an Arithmetic Gradient: Investment

An investment of \$400,000 pays benefits of 20% (\$80,000) in year 1, with an additional \$15,000 added to each previous year's benefit, for 5 years. At a 10% discount rate, is it worth it?

$$\begin{aligned} P &= -400,000 + 80,000(P/A, 10\%, 5) + 15,000(P/G, 10\%, 5) \\ &= -400,000 + 80,000(3.7908) + 15,000(6.8618) \\ &= -400,000 + 406,190 = +\$6,190 \end{aligned}$$

Barely worth it: the benefits' PV just exceeds the \$400,000 cost.

Example Present Value with an Arithmetic Gradient: Maintenance

A machine costs \$155 to maintain this year, with maintenance costing \$35 more each subsequent year, over an 8-year life at 6%. How much must be invested now to cover maintenance, and what is the Equivalent Annual Cost (EAC)?

$$\begin{aligned} P &= 155(P/A, 6\%, 8) + 35(P/G, 6\%, 8) = 155(6.2098) + 35(19.8416) = \$1,656.97 \\ \text{EAC} &= P(A/P, 6\%, 8) = 1,656.97(0.16104) = \$266.83 \text{ per year} \end{aligned}$$

With a \$1,000 overhaul at the start ($t = 0$):

$$\text{EAC} = (1,656.97 + 1,000)(A/P, 6\%, 8) = \$427.87 \text{ per year}$$

Example Future Value with Periodic Dividends

\$100,000 is loaned at 15% (interest compounded annually, not paid out), with a \$5,000 dividend at the end of every third year for 15 years, reinvested at 15%. How much do you have at the end?

Principal: $100,000(1.15)^{15} = \$813,706$. The dividends are an annuity with 3-year periods ($N = 5$), so use the 3-year effective rate $i_3 = (1.15)^3 - 1 = 52.09\%$:

$$\begin{aligned} F &= 100,000(F/P, 15\%, 15) + 5,000(F/A, 52.09\%, 5) \\ &= 813,706 + 68,510 = \$882,216 \end{aligned}$$

Example A Business Example

Over 3 years a small business has:

- sales of \$240k, \$360k, \$480k per year, received in equal monthly amounts (\$20k, \$30k, \$40k/month)
- investments of \$15k/quarter in year 1, \$20k/quarter in year 2, \$25k/quarter in year 3
- an initial investment of \$100k, and a salvage value of \$70k after 3 years

What is the PV at 12%, analysed on monthly cash-flows?

Monthly rate $i = 12\%/12 = 1\%$. Each year's sales are a 12-month annuity shifted to the start of its year; quarterly investments fall at months 3, 6, ..., 36:

$$\begin{aligned} P &= -100,000 + [20,000 + 30,000(P/F, 1\%, 12) + 40,000(P/F, 1\%, 24)](P/A, 1\%, 12) \\ &\quad - \sum_{q=1}^{12} Q_q(P/F, 1\%, 3q) + 70,000(P/F, 1\%, 36) \\ &\approx \$633,463 \end{aligned}$$

where Q_q is \$15k, \$20k or \$25k depending on the year.

Week 3 - Financial Instruments

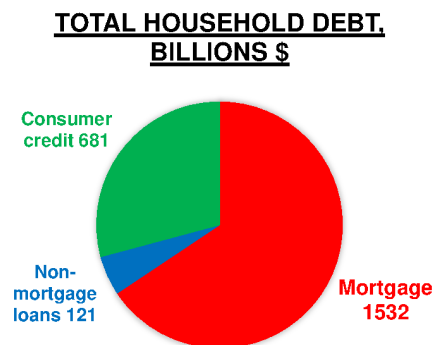
3.1 - Mortgages

Definition - **Mortgage**

A loan secured by **real property**: you borrow money and use the property as collateral. From French law, “death contract” (*mort-* = death): the pledge ends when the obligation is fulfilled.

Importance of Mortgages

- The most common way to finance a home: new purchases with a mortgage outnumber those without by a factor of 6 (CMHC)
- Mortgages are by far the largest share of Canadian household debt



Total household debt, billions \$ (Statistics Canada)

Mortgage Terminology

Buying a \$500,000 house:

- **Principal**: the amount you borrow, up to some fraction of the price (e.g. 80% $\Rightarrow$ \$400,000)
- **Down payment**: the rest, which you pay yourself (\$100,000)
- **Loan-to-value ratio (LTV)**: mortgage loan / value of house = $400,000 / 500,000 = 0.8$
- **Mortgage rate**: the interest rate the bank charges on the principal (e.g. 5%/year)
- **Amortization period**: the agreed time horizon for full repayment (e.g. 25 years)
- **Term**: the shorter period over which the agreed rate is fixed (e.g. 5 years). After the term ends, the rate is renegotiated and the payments may change

Remark. Payments are calculated based on the **amortization period**, **not** the term! Canadian mortgage rates are quoted as nominal annual rates with **semi-annual compounding**, so convert to the equivalent monthly rate before using $(A/P, i, N)$.

Example Mortgage Payments Across Terms

A \$500,000 home with a 20% down payment. The mortgage rate is 5%, with a 5-year term and a 25-year amortization period.

a) What is the monthly payment during this term?

Principal $P = 0.8(500,000) = \$400,000$. Semi-annual rate 2.5%, so the monthly rate satisfies $(1 + i_m)^6 = 1.025$:

$$i_m = 1.025^{1/6} - 1 = 0.41239\%, \quad N = 25 \times 12 = 300$$

$$A = 400,000(A/P, 0.41239\%, 300) = \$2,326.42 \text{ per month}$$

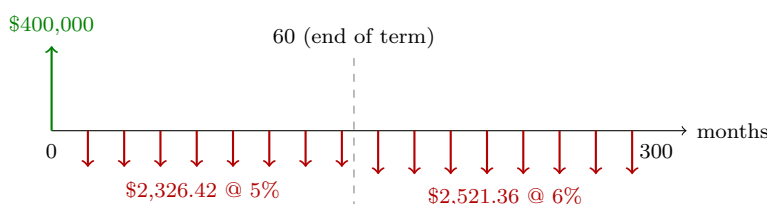
b) After the term, the new mortgage rate is 6%. What is the new monthly payment?

The amount still owed after 60 payments is the PV of the 240 remaining payments at the old rate:

$$B_{60} = 2,326.42(P/A, 0.41239\%, 240) = \$354,030$$

Re-amortize this over the remaining 20 years at the new rate, $i'_m = 1.03^{1/6} - 1 = 0.49386\%$:

$$A' = 354,030(A/P, 0.49386\%, 240) = \$2,521.36 \text{ per month}$$



Borrower's view: payments step up at the renewal

Some Canadian Mortgage Rules

- Since 2012, amortization periods of up to 25 years are allowed
- For 25-year amortization, the down payment must be at least 20%
- For shorter amortization periods, the down payment may be smaller ($< 20\%$), but **mortgage insurance** is required
- Real-life mortgages can be more complicated: rates can be variable within a term, you can pay the mortgage down faster, ...

Summary

1. Mortgages are loans backed by real property (the most common way to finance a home)
2. Payments are calculated from the principal, mortgage rate, and **amortization period**
3. At the end of each term, the next payments are set by the **amount still owed**, the **new rate**, and the **remaining amortization period**

3.2 - Bonds

Definition - **Bond**

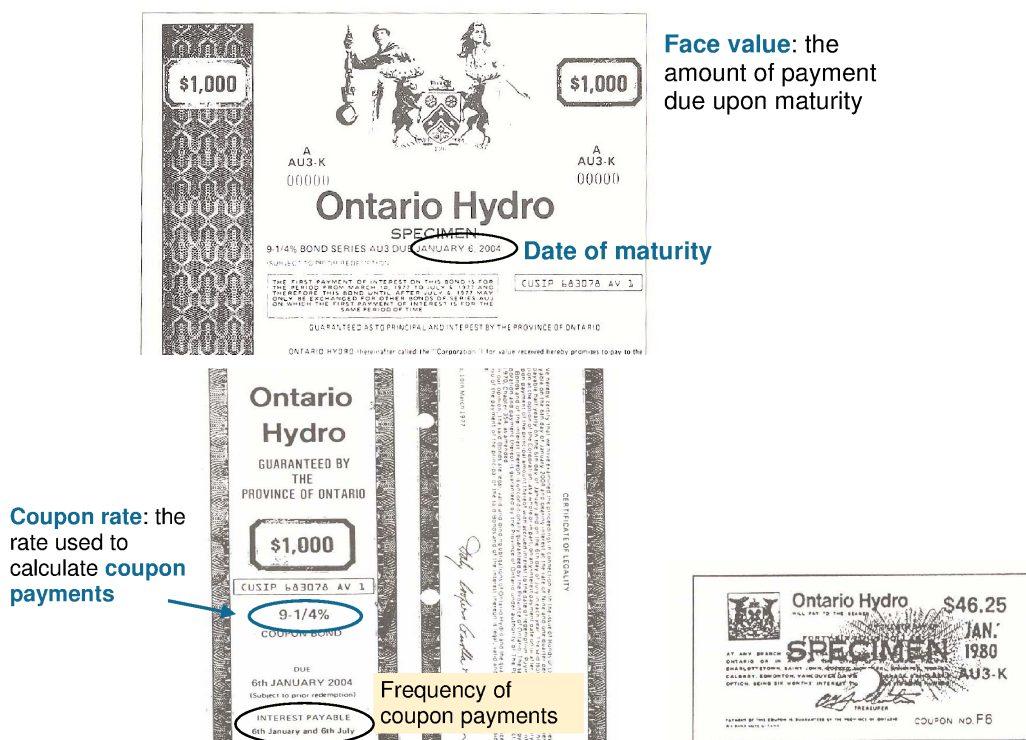
A financial instrument issued by firms and governments (federal, provincial, municipal) to raise funds for projects. It is a special form of loan, usually long-term (up to 30 years). The issuer (the borrower) promises to pay:

- a stated amount at specified intervals for a defined period: the **coupon payments**
- a final amount at a specific date: the **face value**

Bonds are usually tradeable at any time.

Reading a Bond (Ontario Hydro)

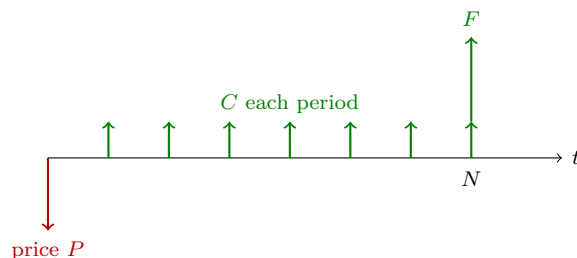
- **Face value**: the amount paid at maturity (\$1,000)
- **Maturity date**: when the face value is paid (January 6, 2004)
- **Coupon rate**: the (nominal annual) rate used to calculate coupon payments ($9\frac{1}{4}\%$)
- **Coupon frequency**: how often coupons are paid (January 6 and July 6, i.e. semi-annually)



Each semi-annual coupon = $1000 \times 9.25\% / 2 = \46.25 , matching the coupon shown.

Bond Cash-flow Diagram

From the bondholder's view: pay the price P now, receive coupon C every period for N periods, plus the face value F at maturity.



$$P = C(P/A, i, N) + F(P/F, i, N)$$

where i is the yield per coupon period.

Example How Much Is This Bond Worth?

Ontario Hydro will pay you \$1,000 in 10 years, plus coupons at 9.25% paid every 6 months. If the bank pays 10% interest, what amount must be deposited now to produce the same cash-flows?

$C = \$46.25$, $N = 20$ half-years, $i = 10\%/2 = 5\%$ per half-year:

$$\begin{aligned} P &= 46.25(P/A, 5\%, 20) + 1000(P/F, 5\%, 20) \\ &= 46.25(12.4622) + 1000(0.37689) = \$953.27 \end{aligned}$$

Would you be willing to pay \$953?

Only if the bond were as safe as the bank. Ontario Hydro might default, so you should demand a lower price.

Definition - Yield

Risk == reward: the buyer must be compensated for the risk of default by paying a **lower** price. The **yield** is the (hypothetical) interest rate that makes the discounted bond payments equal to the price. The yield reflects the time value of money for the bond's payments.

Example Yield from a Price

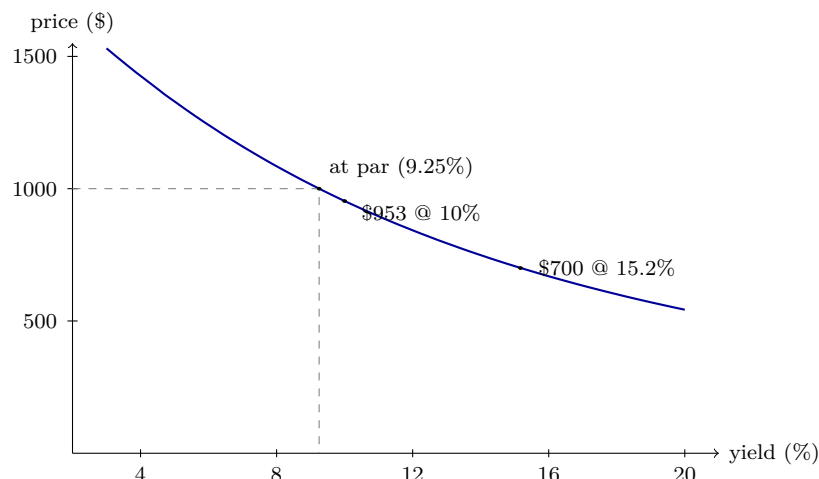
If the highest price you are willing to pay for the bond above is \$700, what annual interest rate gives that price?

$$700 = 46.25(P/A, i, 20) + 1000(P/F, i, 20)$$

Solving numerically (trial and error or a solver): $i = 7.587\%$ per half-year, i.e. a nominal yield of $\approx 15.2\%$ /year (effective 15.75%).

Bond Price Is Determined by Yield

- Price = payments discounted at the yield, so a higher yield means a lower price
- When yield = coupon rate, price = face value ("at par"). When yield > coupon rate, price < face value ("at a discount"), and vice versa



Ontario Hydro bond (\$1,000 face, 9.25% semi-annual coupon, 10 years)

Example Reflection

If the bond yield increases, does the price of the bond increase or decrease?

The price **decreases**: the same payments are discounted at a higher rate.

Yield Depends on Risk and Time to Maturity

Country	2-yr gov't yield (2019, %)	Maturity (yrs)	Gov't of Canada yield (%)
Argentina (1 yr)	32.59	3	1.00
Brazil	5.69	5	1.50
Canada	1.521	7	2.25
China	2.700	10	2.25
Germany	-0.828	Long (30)	2.75
Uganda	16.000	<i>Longer time to maturity $\Rightarrow$ higher yield</i>	
Ukraine	16.97		
United Kingdom	0.449		
United States	1.575		

More stable economies $\Rightarrow$ lower yield

This is the same idea as the Week 1 interest-rate factors: yield rises with credit (default) risk and maturity risk.

Summary

1. **Face value, coupon payments/rate, and maturity** are defined by the bond itself
2. The **yield** is determined by the market (how risky the investment is, interest rate expectations)
3. Bond prices are calculated by discounting the payments at the bond yield

3.3 - Bond Examples

Example Bond Price

A \$10,000 bond maturing in 8 years has a 12% coupon payable quarterly. If the yield is 10%, how much would you pay for the bond?

$C = 10,000 \times 12\%/4 = \300 per quarter, $N = 32$ quarters, $i = 10\%/4 = 2.5\%$ per quarter:

$$\begin{aligned} P &= 300(P/A, 2.5\%, 32) + 10,000(P/F, 2.5\%, 32) \\ &= 300(21.8492) + 10,000(0.45377) = \$11,092.46 \end{aligned}$$

Price > face value, because the coupon rate (12%) exceeds the yield (10%).

Example Yield on a Bond

You buy a bond for \$1,000. The issuer pays a \$45 coupon every 6 months and a face value of \$880 at the end of 2 years. What is the **effective annual** yield?

Solve for the semi-annual rate i with $N = 4$:

$$1000 = 45(P/A, i, 4) + 880(P/F, i, 4) \Rightarrow i = 1.570\% \text{ per half-year}$$

$$i_e = (1 + 0.01570)^2 - 1 = 3.16\%$$

The yield is low despite the 9%/year coupons, because you only get \$880 back on a \$1,000 purchase.

Week 4 - Risk, Reward and Arbitrage

4.1 - Capital Asset Pricing Model

Definition - **Valuation**

The analytical process of determining the current (or projected) worth of an asset or a company. A valuation has two components:

- **Projected cash-flows**
- **Interest rate:** determined by risk (uncertainty)

Engineers often need project valuations to make decisions, e.g. lease or buy, replace or keep current equipment, go or no-go on a project.

Why Proper Valuations Are Important

Auger and Guzman (2010) looked at 51 copper-mining investments (1957–1999):

- Half of the projects were invested in at a sub-optimal time
- 36 projects should have been designed with a 40% larger or smaller capacity
- Sub-optimal decisions led to a total NPV **49% lower** than optimal

Conclusion: use better valuation methods (Real Options Analysis).

Risk versus Reward

You have \$10,000 to buy a car after graduation (2 years from now). Where do you put it? Bank account, government bond, corporate bond, a stock, a startup, a loan to an acquaintance. . . Going down this list, the payoff gets more uncertain, so you would demand a higher expected return.

Definition - **Stock Market (“The Market”)**

The aggregation of buyers and sellers of **stocks** (shares), which represent ownership claims on businesses. Only stocks of “public” companies are traded, usually through brokerages and electronic trading platforms.

Definition - **Financial Risk**

The **uncertainty in a future payoff**.

Example Two-State Future

Only two future states are possible. Asset A pays \$100 in either state; Asset B pays \$120 in state 1 and \$70 in state 2.

A is risk-free, so we discount it at the risk-free rate. B is risky: what rate should we use? (Answered by replication in Topic 4.2.)

Project or Company Returns

With prices P_{t_j} observed every Δt , the (continuously compounded) return over each period is

$$P_{t_j} = P_{t_{j-1}} e^{r_{t_j} \Delta t} \iff r_{t_j} = \frac{1}{\Delta t} \ln \frac{P_{t_j}}{P_{t_{j-1}}}$$

Stack the returns of the i -th stock into a vector $\vec{R}_i = [r_{t_1}, r_{t_2}, \dots, r_{t_m}]^T$ and define

$$\sigma_i^2 = \text{Var}(\vec{R}_i), \quad \text{Volatility} = \sigma_i$$

Example Returns from Prices

t	0	1	2	3	4
P	20	21	22	20	21
$r_t (\Delta t = 1)$	–	0.0488	0.0465	–0.0953	0.0488

Mean return $\approx 1.22\%$ per period, volatility $\sigma \approx 0.0717$ (sample std. dev.).

Modern Portfolio Theory (MPT)

- Introduced by Harry Markowitz (1952)
- An investment's risk and return should not be viewed alone, but by how it affects the **overall portfolio's** risk and return
- MPT gives a mathematical framework for building a portfolio that **maximizes expected return for a given level of risk**

Invest a fraction x_i of wealth in the i -th asset:

$$\vec{X} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1n} \\ \vdots & \ddots & \vdots \\ \sigma_{n1} & \cdots & \sigma_{nn} \end{bmatrix}, \quad \sigma_{ii} = \text{Var}(\vec{R}_i), \quad \sigma_{ij} = \sigma_{ji} = \text{Cov}(\vec{R}_i, \vec{R}_j)$$

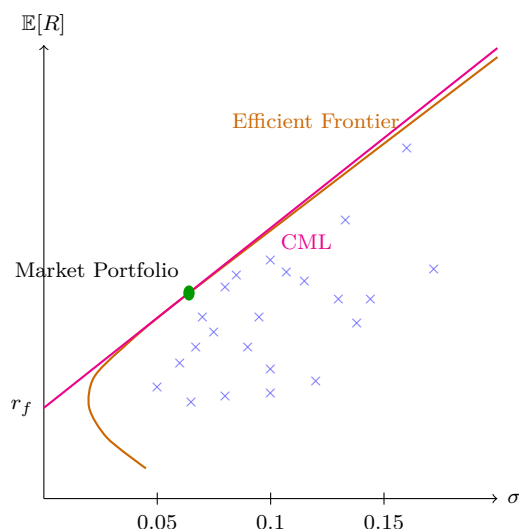
$$\mathbb{E}[R_p] = x_1 \mathbb{E}[R_1] + \cdots + x_n \mathbb{E}[R_n], \quad \sigma_p = \sqrt{\vec{X}^T \Sigma \vec{X}}$$

Optimize: maximize $\mathbb{E}[R_p]$ for a given σ_p .

CAPM Assumptions

- Correlations between assets are fixed and constant forever
- All investors maximize economic utility (make as much money as possible) and are rational and risk-averse
- All investors have the same information at the same time, and their probability beliefs match the true distribution of returns
- No taxes or transaction costs
- All investors are **price takers** (their actions do not influence prices)
- Anyone can lend and borrow unlimited amounts at the risk-free rate
- All securities can be divided into parcels of any size

Efficient Frontier and Capital Market Line



- **Efficient frontier**: the best expected return achievable for each volatility using portfolios of risky assets
- **Capital market line (CML)**: the line from r_f tangent to the frontier. The tangent point is the **market portfolio (MP)**
- Mixing the MP with lending/borrowing at r_f reaches any point on the CML, which beats the frontier everywhere else

Example Leveraging Up the CML

$r_f = 3\%$, $\mathbb{E}[R_{MP}] = 7\%$, $\sigma_{MP} = 0.06$. Invest \$100 in the MP, then borrow another \$50 at r_f and invest it in the MP too.

You hold $1.5\times$ your wealth in the MP:

$$\sigma_p = 1.5 \sigma_{MP} = 0.09, \quad \mathbb{E}[R_p] = 1.5(7\% - 3\%) + 3\% = 9\%$$

Systematic vs. Idiosyncratic Risk

Regress a company's return on the market's:

$$R_{C,t} = \alpha_C + \beta'_C R_{MP,t} + \varepsilon_{C,t}$$

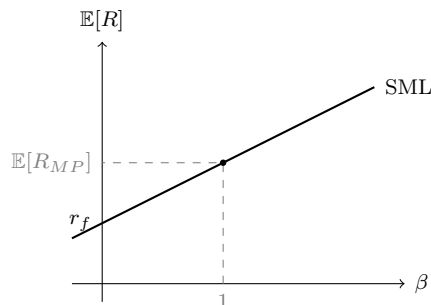
- Volatility in $R_{MP,t}$ is **systematic** ("market") risk
- Volatility in $\varepsilon_{C,t}$ is **idiosyncratic** ("firm-specific") risk
- Investors can "diversify away" idiosyncratic risk with portfolios, so there is **no reward** for bearing it. Investors are only rewarded for market risk, measured by β

Definition - CAPM

The expected return of the i -th asset is

$$\boxed{\mathbb{E}[R_i] = r_f + \beta_i (\mathbb{E}[R_{MP}] - r_f)}, \quad \beta_i = \frac{\sigma_{i,MP}}{\sigma_{MP}^2} = \frac{\rho_{i,MP} \sigma_i}{\sigma_{MP}}$$

$\mathbb{E}[R_{MP}] - r_f$ is the **market risk premium**. The MP is often estimated with a stock index.



Security market line: intercept r_f , slope $\mathbb{E}[R_{MP}] - r_f$

Example Estimating Stock Return

Market: $r_f = 2\%$, $\mathbb{E}[R_{MP}] = 8\%$, $\sigma_{MP} = 10\%$. Company X: correlation to the market 0.6, volatility 0.15. Estimate X's expected return.

$$\beta_X = \frac{0.6 \times 0.15}{0.10} = 0.9$$

$$\mathbb{E}[R_X] = 2\% + 0.9(8\% - 2\%) = 7.4\%$$

(The handwritten slide used 10% for the market return, giving 9.2%. With the stated 8% it is 7.4%.)

Example Estimating Future Stock Price

A company with $\beta = 0.9$ trades at \$170/share. $r_f = 2\%$, market rate = 7%. What is the expected share price in 5 years if it pays no dividends?

$$\mathbb{E}[R_C] = 2\% + 0.9(7\% - 2\%) = 6.5\%$$

$$P_5 = P_0 e^{\mathbb{E}[R_C]t} = 170e^{0.065 \times 5} = \$235$$

Example Reflection: Stock with Dividends

A stock trades at \$10 and pays quarterly dividends of \$0.10/share. $\mathbb{E}[R_{MP}] = 10\%$, $r_f = 3\%$, $\beta = 1.2$. What is the expected price 3 years from now?

$$\mathbb{E}[R_C] = 3\% + 1.2(10\% - 3\%) = 11.4\%/yr$$

Dividends are quarterly, so convert: $(1 + i_q)^4 = 1.114 \Rightarrow i_q = 2.736\%$. Today's price is the PV of 12 dividends plus the price at year 3:

$$10 = 0.10(P/A, 2.736\%, 12) + P_3(P/F, 11.4\%, 3) \Rightarrow P_3 = \$12.43$$

4.2 - Arbitrage

Definition - Arbitrage

- **In financial markets:** taking advantage of a price difference between two or more markets
- **Academic use:** an opportunity to achieve a **risk-free** gain at a rate **greater than the risk-free rate**
- **Statistical arbitrage:** a gain at a higher rate than one should get for a given level of risk

Fundamental to finance theory (and to valuation) is the **no-arbitrage assumption**: “no free lunch”.

Example Corn Arbitrage

You can buy a bushel of corn for \$10 today and enter a forward contract to deliver it for \$12 one year from now (storage is free). You can also borrow at 10%. What would you do?

Borrow \$10 x and buy x bushels today: net cash-flow \$0. In one year, sell the x bushels for \$12 x and repay the \$11 x loan: a risk-free profit of \$1 x .

As $x \rightarrow \infty$ the profit is infinite. In reality, demand for corn today pushes the price up until the arbitrage disappears. The no-arbitrage price is

$$\frac{\$12}{1.10} = \$10.91$$

Futures and Forwards

- Contracts used to **hedge** against risks or to **speculate**
- Examples of **derivative** assets: their value derives from an underlying asset
- Both lock in a specific price for a certain asset

Definition - Forward Contract

An obligation to buy or sell a certain asset at a specified price (the **forward price**) at a specified time (the **maturity** or expiration date). Typically not traded on exchanges; both buyer and seller are obligated to fulfill their end at maturity.

Definition - Futures Contract

The same as a forward, except futures are **settled daily** (so they can be bought or sold at any time) and are traded on a **standardized exchange**.

Futures	Forwards
Settled daily	Settled at maturity
Standardized	Not standardized
Low risk of non-fulfilment (regulated)	Low regulation/oversight on settlement
Traded on public exchanges	Private contract between two parties

E.g. the WTI oil futures curve: the price today for oil delivered in month 1, 2, ... (in March 2020 the curve rose steeply from \$25 to \$45).

Example Interest Rate Forward

Given a yield curve $r(t)$, find the forward rate $r_{1,2}$ that one could lock in today for borrowing/lending from t_1 to t_2 .

Invest P_0 today. Directly to t_2 : $F_2 = P_0 e^{r_{0,2}t_2}$. Or to t_1 then roll at the forward rate: $F'_2 = P_0 e^{r_{0,1}t_1} e^{r_{1,2}(t_2-t_1)}$. No arbitrage requires $F'_2 = F_2$:

$$r_{1,2} = \frac{r_{0,2}t_2 - r_{0,1}t_1}{t_2 - t_1}$$

Example Foreign Exchange Forward

Canadian risk-free rate 3%, US risk-free rate 2.5%, current FX = 0.9 US\$/C\$. What is the fair one-year forward FX rate?

Two risk-free ways to turn C\$1 today into money in one year must agree:

- Keep it in C\$: $1 \times 1.03 = \text{C}\1.03
- Convert to US\$0.90, invest: $0.90 \times 1.025 = \text{US}\0.9225

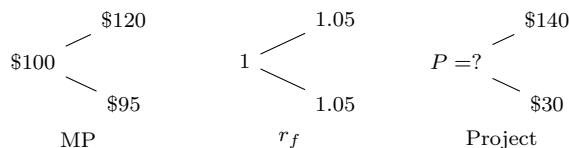
$$FX_1 = \frac{\text{US}\$0.9225}{\text{C}\$1.03} = 0.896 \text{ US}\$/\text{C}\$$$

Using Replication to Price an Asset

Build a portfolio of assets we can already price that has the **same payoff in every state**. By no-arbitrage, the asset's price equals the portfolio's price.

Example Pricing a Project by Replication

A project pays \$140 if the market goes up and \$30 if it goes down. The risk-free rate is 5% for the period; the MP is \$100 today and will be worth \$120 (up) or \$95 (down). What is a fair price for the project?



Hold a units of the MP and b units of the risk-free asset:

$$\text{Up: } 120a + 1.05b = 140$$

$$\text{Down: } 95a + 1.05b = 30$$

$\Rightarrow a = 4.4, b = -369.52$ (borrow).

$$P = 100(4.4) - 1(369.52) = \$70.48$$

Note that we did **not** use the probabilities!

Example Expected Returns and the Project's Beta

Same setup; the probability the market goes up is 60%.

$$\mathbb{E}[R_{MP}] = \frac{120(0.6) + 95(0.4)}{100} - 1 = 10\%$$

$$\mathbb{E}[R_{proj}] = \frac{140(0.6) + 30(0.4)}{70.48} - 1 = 36.2\%$$

Back out the project's beta from CAPM:

$$36.2\% = 5\% + \beta_{proj}(10\% - 5\%) \Rightarrow \beta_{proj} = 6.24$$

The project swings far more than the market, so it is heavily “leveraged” market risk.

Example Payoffs Independent of the Market

- a) What if the project paid \$140 regardless of what the market did?

It is risk-free: $P = 140/1.05 = \$133$.

- b) What if the payoff had a 50% chance of being \$150 and 50% of being \$130 (independent of the market)?

$\mathbb{E}[\text{payoff}] = 0.5(150) + 0.5(130) = \140 . The risk is purely idiosyncratic (diversifiable), so it carries no premium: $P = 140/1.05 = \$133$.

Example Systematic and Idiosyncratic Risk

A wealthy (risk-neutral) investor can invest in a product development project. There is a 60% chance the technology works very well (independent of the market). Payoffs:

	Tech works (60%)	Doesn't (40%)
Market up	\$80k	\$50k
Market down	\$55k	\$20k

The MP is \$45 today, worth \$60 (up) or \$40 (down); $r_f = 5\%$. An initial R&D investment of \$20k is required. What is the maximum price for the project?

The technology risk is idiosyncratic, so replace it by its expectation in each market state:

$$\mathbb{E}[P_u] = 0.6(80k) + 0.4(50k) = 68k, \quad \mathbb{E}[P_d] = 0.6(55k) + 0.4(20k) = 41k$$

Replicate the market risk:

$$60a + 1.05b = 68k$$

$$40a + 1.05b = 41k$$

$$\Rightarrow a = 1.35k, b = -12.38k.$$

$$P_0 = 1.35k(45) - 12.38k = \$48.37k$$

Since \$20k of R&D is also needed, buy the project for **\$28.37k or less**.

Example Bond: Accounting for Default

A company issues a 5-year bond (face \$100) with a 5% annual coupon. The **risk-neutral probability** of default in any given year is 5%, and on default only half of the principal (\$50) is paid out. The risk-free rate is 3%.

- a) What is the probability of default over the 5 years?

$$1 - 0.95^5 = 22.6\%$$

- b) What should the price of the bond be?

Work backwards through the tree. Each year the bond either survives (95%, pays the coupon plus the value going forward) or defaults (5%, pays \$50). Discount each step at r_f since these are risk-neutral probabilities:

$$\begin{aligned}P_{up}^{(4)} &= 5 + \frac{0.95(105) + 0.05(50)}{1.03} = 104.27 \\P_{up}^{(3)} &= 5 + \frac{0.95(104.27) + 0.05(50)}{1.03} = 103.60 \\P_{up}^{(2)} &= 5 + \frac{0.95(103.60) + 0.05(50)}{1.03} = 102.98 \\P_{up}^{(1)} &= 5 + \frac{0.95(102.98) + 0.05(50)}{1.03} = 102.41 \\P_0 &= \frac{0.95(102.41) + 0.05(50)}{1.03} = \textbf{\$96.88}\end{aligned}$$

- c) What is the yield?

$$96.88 = 5(P/A, y, 5) + 100(P/F, y, 5) \Rightarrow y = \textbf{5.735\%}$$

The yield is above the 3% risk-free rate (and the 5% coupon) to compensate for default risk.

Summary

1. Risk is the uncertainty in a future payoff; investors demand a higher expected return for bearing it
2. Only **systematic** (market) risk is rewarded. CAPM: $\mathbb{E}[R_i] = r_f + \beta_i(\mathbb{E}[R_{MP}] - r_f)$
3. **No arbitrage**: two ways to get the same payoff must cost the same. This prices forwards (corn, interest rates, FX)
4. **Replication**: match an asset's payoffs with the MP and the risk-free asset; the price does not need the real-world probabilities